Our M1 Plan and Reproduction Results

1. Our frozen M1 plan

Research question

Can we reproduce the paper's finding that negative comments affect how urgent developers think a problem is, using the authors' original data and code?

Sample

We used data from 59 developers, who each rated the urgency of four code comments. This gave us 236 ratings in total, with each developer rating two neutral comments and two negative comments.

The developers were split into three groups based on their answer to the survey question: "Does a negative comment make a problem seem more important to you?"

- 22 agreed.
- 11 had no opinion.
- 26 disagreed.

Method

We ran the authors' original model on the urgency ratings using a fixed random seed. This allows us to repeat the analysis with the same random settings.

Metric

We calculated an odds ratio for each group to compare how often negative comments were rated as more urgent rather than less urgent. An odds ratio of 1 means there was no difference.

Decision rule

Before running the experiment, we decided that the result would count as reproduced if the agree group had an odds ratio between 1.85 and 2.05, with strong evidence. The other two groups also needed to have values close to 1.

2. Our execution procedure

We used two separate approaches to check the results.

Main run

We ran the authors' notebook, work/analyse_urgency_seeded.Rmd, inside Docker. We used the authors' original software setup, with one small fix to make it work.

We used a single command to run the notebook and save the results.

Independent cross-check by Saif

Saif wrote a separate script called output_urgency/run_urgency.R on the saif branch, at commit 283d458.

His script runs the same model on the same urgency data, but it does not use the authors' notebook. Instead, it uses his own script, a different random seed (4928), a different random-number setting and a modified Docker setup.

This gives us a second way to check whether the main result holds when the analysis is run differently.

3. Our raw evidence

Main run

- work/analyse_urgency_seeded.html contains the full notebook output.
- work/Urgency perception-1.csv, work/Urgency perception-2.csv and work/Urgency perception-3.csv contain 100,000 simulated results for each group.
- work/evidence Urgency.csv contains the strength-of-evidence values.

Saif's independent run

- output_urgency/Urgency perception-1.csv, output_urgency/Urgency perception-2.csv and output_urgency/Urgency perception-3.csv contain 100,000 simulated results for each group.
- output_urgency/urgency_posterior_summary.csv contains his summary results, including a strength-of-evidence value of 113.29 for the agree group.

These files allow us to inspect the results from both runs and check how they were produced.

4. Tracing our results

We can trace the main result of **1.94 for the agree group** back to the original data and code.

1. The result is recorded in RESULT.md.
2. It comes from work/Urgency perception-3.csv, where 48,741 results were classified as more urgent and 25,121 as less urgent. The ratio is approximately 1.94.
3. The file was produced by running analyse_urgency_seeded.Rmd with the random seed 20261013.
4. The notebook read the authors' original data file, export_jan_23_no_ip_no_email.csv, which contains ratings from 59 developers across four comments each.

Saif's result can also be traced back to its source:

1. His agree-group result was approximately 1.93.
2. It came from 48,671 results classified as more urgent and 25,184 as less urgent in output_urgency/Urgency perception-3.csv.
3. The file was generated by running output_urgency/run_urgency.R with seed 4928.

This gives us two sets of results that can be checked against their respective execution procedures and the original input data.

5. Changes made to the original setup

Changes to our main run

We made four changes, all recorded in DEVIATIONS.md before running the experiment:

1. Added a line to fix an installation problem with the authors' software setup.
2. Set a fixed random seed so the results could be repeated.
3. Ran only the urgency analysis because that was the part relevant to our research question.
4. Saved the evidence strength values to a file because the original code only printed a general description, such as "Strong".

These changes did not alter the data, model or analysis method.

Differences in Saif's run

Saif's run is separate from our main run, so it is not covered by those four changes. His approach differs from the authors' setup because he used his own script, seed 4928, a different random-number setting and a modified Docker setup.

These differences are described in RESULT.md, but they have not yet been recorded in a formal deviations file.

For that reason, we treat Saif's results as a supporting cross-check rather than the official reproduction result. We also only inspected his output files in Git; we did not run his script ourselves.

6. Current reproduction status: REPRODUCED

Our main run met the conditions we set before running the experiment.

The agree group had an odds ratio of **1.94**, which falls within our target range of 1.85–2.05 and had strong evidence. The other two groups had values of 1.06 and 1.09, both close to 1.

We also repeated the main analysis with three different random seeds. These runs produced values of 1.95, 1.96 and 1.96, which are very similar to the original result.

Saif's separate analysis reached a similar conclusion. His results were:

- Agree group: 1.93, with strong evidence of 113.29.
- No-opinion group: 1.04.
- Disagree group: 1.09.

The fact that two different approaches produced similar results gives us more confidence that the finding is stable and is not simply a result of one particular script or random seed.

However, there is one limitation: we inspected Saif's output files but did not independently rerun his script. His results therefore support our conclusion, but we have not fully verified his execution ourselves.

Based on our main run and the decision rule established in M1, our current status is **REPRODUCED**.

7. Held-out data

Both our main run and Saif's script used only the material originally released to our group, including the authors' data, code and setup files.

We have not opened or used any Event or held-out material, so we have stayed within the limits of our M1 plan.